

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	01 October 2026
Team ID	6ab6292ddcdae33acb3906da
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The architecture diagram below shows how spreadsheet data is imported into ServiceNow with a transform map and how access to it is controlled. Table 1 and Table 2 give the components and application characteristics.

Project: Importing employee training data into ServiceNow using Import Sets and Transform Maps

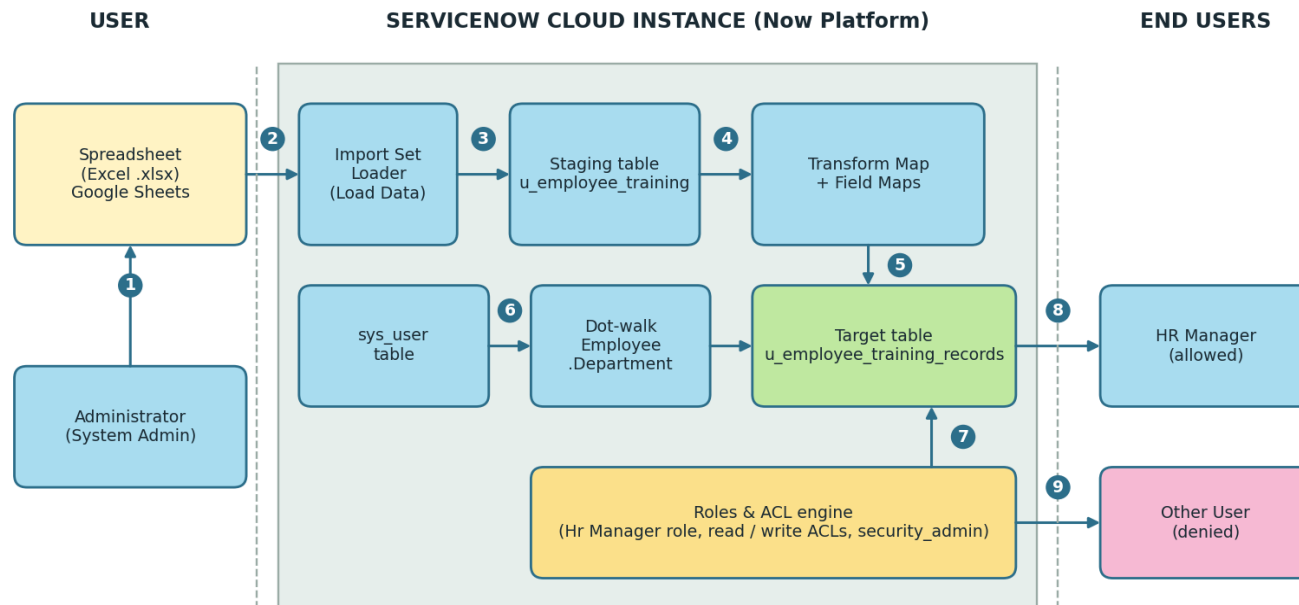


Figure 1: Technical architecture

Numbers in the diagram

1. Administrator prepares the .xlsx file with Training Name, Completion Date, Status and Employee.
2. The file is uploaded through Load Data (Import Set Loader).
3. Rows are staged in the import set table u_employee_training.
4. The transform map and its field maps process the staged rows.
5. Records are inserted into the target table u_employee_training_records.
6. Dot-walking reads Employee.Department from the sys_user table.
7. The Roles & ACL engine checks the user's roles on every request.
8. A user with the Hr Manager role can view and edit the records.
9. Any other user is denied access to the table.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Users work through the ServiceNow web UI: forms, list views, application menu and module for Employee Training Records, and the Impersonate User option for testing.	ServiceNow Next Experience (Platform UI) in a web browser
2.	Application Logic-1 (Data import)	Load Data uploads the .xlsx file (sheet 1, header row 1) and creates the import set table u_employee_training; rows are staged before they reach the real table.	ServiceNow Import Sets (Load Data, ImportProcessor)
3.	Application Logic-2 (Transform)	The transform map links the import set table to the target table; field maps connect u_training_name, u_completion_date, u_status and the employee column. Running the transform inserts the records.	ServiceNow Transform Map and Field Maps
4.	Application Logic-3 (Dot-walking)	The Employee reference field links each record to sys_user. Employee.Department is added to the form layout so the department shows on the form and in the list.	ServiceNow Form Layout (dot-walking)
5.	Application Logic-4 (Access control)	The Hr Manager role is created and assigned to a user, added to the application menu and module, and used in read and write ACLs on the table.	ServiceNow Roles, Users and Access Control (ACL)
6.	Database	Tables: u_employee_training (staging), u_employee_training_records (target; fields Training Name, Completion Date, Status [choice], Employee [reference to sys_user]) and sys_user.	ServiceNow relational tables (Table & Dictionary entries)
7.	Cloud Database	The tables above live in the database of the ServiceNow cloud instance, which ServiceNow manages. No separate database server is set up.	ServiceNow cloud instance database
8.	File Storage	The spreadsheet is created and downloaded as .xlsx, then uploaded to the instance through Load Data. No other file storage is required.	Google Sheets / Excel (.xlsx), ServiceNow attachments
9.	External API-1	Not used. Data comes only from the uploaded spreadsheet.	Not applicable
10.	External API-2	Not used.	Not applicable
11.	Machine Learning Model	Not used. The project is rule-based (transform map, roles, ACLs).	Not applicable
12.	Infrastructure (Server / Cloud)	Local server configuration: none; only a browser and a spreadsheet tool are needed. Cloud server configuration: a ServiceNow cloud instance (application scope: Global).	ServiceNow cloud instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source frameworks are used. The project is built with the low-code ServiceNow platform. The spreadsheet uses the open .xlsx file format.	ServiceNow low-code platform; Office Open XML (.xlsx)
2.	Security Implementations	Role-based access: the Hr Manager role is required in read and write ACLs on u_employee_training_records and on the application menu and module. ACL changes need the security_admin role (elevated). Access was tested by impersonating users.	ServiceNow Roles, ACLs, Elevate Role, Impersonate User
3.	Scalable Architecture	Importing is split into stages (spreadsheet, staging table, target table), and the same transform map can be reused for later loads. The platform runs as a multi-tier cloud service (browser, application, database).	Import Sets and Transform Maps on the ServiceNow cloud
4.	Availability	The application runs on the ServiceNow-hosted cloud instance, so servers, backups and uptime are handled by the platform and nothing is installed locally.	ServiceNow managed cloud instance
5.	Performance	Staging lets the file load quickly before transformation. Department is read through dot-walking, so it is not copied into each record. The Employee reference points to the indexed sys_user table.	Import Set staging; dot-walking; reference field

References:

<https://docs.servicenow.com>
<https://developer.servicenow.com>
<https://c4model.com/>